

Scenario S3-O1-I1 — Our Level 1 (Bootstrapped) / Investor Level 1 (Bootstrapped)

Scenario Summary

Maximum-constraint early-investor path. The founders self-fund at bootstrapped intensity (~~40% of ideal spend~~) through ~~Pre-Pilot and Pilot 1 only~~ (5-7 months), then a bootstrapped-level investor arrives right after Pilot 1 completes (approximately M7-M8). Both parties operate at bare-minimum budgets throughout. The earlier investor entry compared to S2 reduces founder capital exposure significantly (~\$21,000-32,000 vs. \$36,000-50,000 in S2-O1-I1), but double-bootstrapped budgets still create the longest S3 timeline and thinnest margins.

- **Economic cost to Production:** \$105,000-135,000 / INR 8,715,000-11,205,000
- **Target timeline to Production:** 26-32 months
- **Founder/company capital before investor:** \$21,000-32,000 / INR 1,743,000-2,656,000
- **Investor capital modeled:** \$84,000-103,000 / INR 6,972,000-8,549,000
- **Equity structure:** 60 / 20 / 20 (founder / founding team / investor pool)
- **Main risk to watch:** bootstrapped investor capital barely covers Pilot 2 + Production at minimum staffing; if revenue underperforms, runway evaporates before break-even

Timeline

Phase	Timing	Duration	Minor Checkpoints	Gate to Advance
Pre-Pilot	M0-M5	5 months	Architecture freeze (M2), APK backlog lock (M3), doctor training draft (M4), internal rehearsal cycles (M4-M5).	Core web flows stable at reduced scope, internal rehearsals pass, doctor and admin runbooks versioned.
Pilot 1	M5-M7	2 months	Warm-network onboarding, first paid consults, investor pitch deck finalized, term sheet negotiation begins.	Ethiopia pilot runs live for 2 months with measured throughput; investor term sheet signed by M7.
Pilot 2	M7-M16	9 months	Investor capital deployed (M8), Kenya prep (M12), partner reporting (M13), roster planning, staffing calibration.	5k-10k user-capable operations with stronger APK, cleaner reporting, staffed callback coverage.
Production Readiness	M16-M26	10 months	Compliance closeout (M19), release freeze (M22), launch playbooks (M24), certification refreshers, production support schedule.	Current prototype scope is production-safe across web and patient mobile with documented fallback operations.

Feature and Output Checklist

Phase	Output Checklist
Pre-Pilot	Web core hardened at reduced pace; APK build underway (sequenced, not parallel); doctor/admin training materials drafted; support desk scripts and callback trees defined. Slower iteration due to constrained hours.
Pilot 1	Patient app covers core journeys (registration, dashboard, basic consult); provider web is stable for GP flow; doctor roster, admin callbacks, and manual fulfilment visible in-system. Feature set narrower than L2/L3 — specialist and diagnostics flows deferred. Investor pitch deck finalized with live Pilot 1 data.
Pilot 2	Core patient features complete; partner dashboards functional; staffing ladder active at minimum levels; reporting and payment loops operational but manual-heavy. Kenya prep begins late in phase.
Production Readiness	Current prototype scope is production-grade across web and patient mobile. Trained ops staff with manual fallback available. Compliance documentation complete. Extended timeline allows thorough but slow hardening.

Services and SLAs

Phase	Uptime Target	Response Commitment	Operating Model
Pre-Pilot	95.0%	Next business day	3 core team at reduced utilization + 7 part-time developers at ~40% capacity + 2 advisors; no employed clinical coverage, only training and rehearsals.
Pilot 1	96.0%	Same-day for critical issues	Minimum employed coverage: 2 doctors, 2 admin ops, 1 technical support. Friendly clinics supplement. Lean ops given 2-month window.
Pilot 2	97.5%	4-hour P1 response	Minimum employed coverage: 3 doctors, 3 admin ops, 2 technical support. Callback and partner reporting loops active.
Production	98.5%	2-hour P1 response	Minimum employed coverage: 5 doctors, 5 admin ops, 3 technical support for near-24/7 coverage.

Cost Breakdown

Summary

Category	Amount
Capital expenditure through Production	\$46,200 / INR 3,834,600
Operating expenditure before Production launch	\$72,300 / INR 6,000,900
Total economic spend to Production	\$118,500 / INR 9,835,500

Workstream Detail

Workstream	Economic Cost (USD)	Economic Cost (INR)	Notes
Product and founder office	\$55,440	INR 4,601,520	3 core team at \$20/hr x 80 hrs/mo x 40% utilization adjusted = ~\$2,112/mo pre-investor (5 productive months); post-investor ramp to ~\$2,640/mo for 21 months.
Web platform hardening	\$14,900	INR 1,236,700	745 hours at \$20/hr senior rate. Stretched over longer timeline due to constrained hours.
Patient APK build	\$7,200	INR 597,600	Fixed Android quote: 480 hours at \$15/hr. Sequenced rather than parallel with web work.
QA, DevOps, security	\$6,600	INR 547,800	~330 hours at \$20/hr. Reduced scope; minimum viable hardening.
Admin and care operations	\$82,460	INR 6,844,180	Employed doctors, admin callback staff, and technical support ladder across 21 months of live operations (Pilot 1 through Production).
Partner onboarding and provider success	\$6,200	INR 514,600	Doctor/admin training protocols, SOP writing, partner reporting. Lean execution.
Legal, accounting, compliance	\$3,600	INR 298,800	Friendly-network pricing at economic value.
Infrastructure, phones, and tools	\$21,600	INR 1,792,800	Hosting, devices, telecom/call costs, operational tooling across 26 months. Starts minimal, scales with phases.
Marketing, travel, rehearsals	\$6,000	INR 498,000	Bare minimum. Word-of-mouth and warm-network driven.
Contingency and buffer	\$25,000	INR 2,075,000	~21% buffer on base costs. Essential given bootstrapped constraints and long timeline.

Operational Staffing Cost by Phase

Phase	Duration	Monthly Ops Cost (USD)	Monthly Ops Cost (INR)	Phase Total (USD)
Pre-Pilot (M0-M5)	5 months	\$0	INR 0	\$0
Pilot 1 (M5-M7)	2 months	\$1,959	INR 162,597	\$3,918
Pilot 2 (M7-M16)	9 months	\$3,458	INR 287,014	\$31,122
Production Readiness (M16-M26)	10 months	\$5,240	INR 434,920	\$52,400
Total operational staffing				\$87,440

Infrastructure Cost by Phase

Phase	Duration	Monthly Infra (USD)	Monthly Infra (INR)	Phase Total (USD)
Pre-Pilot	5 months	\$80	INR 6,640	\$400
Pilot 1	2 months	\$250	INR 20,750	\$500
Pilot 2	9 months	\$480	INR 39,840	\$4,320
Production Readiness	10 months	\$800	INR 66,400	\$8,000
One-time equipment (phones, laptops)	—	—	—	\$1,000
Total infrastructure				\$14,220

Effort and Team

Function	Staffing Pattern	Notes
Core team	3 part-time members @ ~4 hrs/day each, ~40% effective utilization pre-investor	Single owner/founder + 2 founding members. Heavy sequencing of tasks.
Technical delivery (pre-investor, M0-M7)	7 part-time developers, modeled at ~200 productive hrs/mo	~40% of ideal 480 hrs/mo. High sequencing, minimal parallelism.
Technical delivery (post-investor, M8-M26)	7 part-time developers, modeled at ~240 productive hrs/mo	Investor capital enables slightly higher utilization (~50% of max). Still constrained.
Advisory board	2 fractional advisors @ ~10-15 hrs/mo each	Finance, compliance, strategic review.
Employed care coverage	Pilot 1: 2 doctors / Pilot 2: 3 / Production: 5	Partner clinics critical at bootstrapped levels. Lower ladder than S2 due to earlier but smaller investor.
Employed admin ops	Pilot 1: 2 admins / Pilot 2: 3 / Production: 5	Callbacks, exception handling, manual fulfilment.
Technical support	Pilot 1: 1 / Pilot 2: 2 / Production: 3	L1/L2 issue triage and live-ops monitoring.

Monthly Team Economic Cost

Period	Dev Team (USD)	Dev Team (INR)	Notes
Pre-investor (M0-M7)	~\$5,930	~INR 492,190	~40% of full \$14,830/mo rate card. Bare minimum.
Post-investor (M8-M26)	~\$7,400	~INR 614,200	~50% of full rate card. Investor enables modest increase.

Revenue and Pricing

Revenue Source	Pilot 1	Pilot 2	Production
GP consults	Subsidised, Ethiopia-first, ~\$2-3 per paid interaction	Full Ethiopia pricing + Kenya tests	Core revenue line
Specialist consults	Minimal	Grows slowly after referral flow stabilizes	Lower-volume margin contributor
Pharmacy commission	Small, admin-assisted	Repeatable, manual-heavy	Scales with partner conversion
Diagnostics commission	Small, admin-assisted	Improves with order flow consistency	Retention lever
Travel/care coordination	Rare, high-touch	Low-volume, higher-value	Strategic differentiator

Revenue Line	Ethiopia Benchmark	Kenya Benchmark	Notes
GP consult	\$3.5 / ETB 470	\$5 / KES 650	Starts subsidised in Pilot 1; reaches benchmark by late Production.
Specialist consult	\$9 / ETB 1,250	\$12 / KES 1,560	Pilot 2 onward.
Pharmacy take-rate	10% of order value	10% of order value	Pilot 1 onward with manual fulfilment.
Diagnostics take-rate	12% of order value	12% of order value	Pilot 1 onward with admin/lab coordination.
Travel/care coordination	\$10-15	\$10-15	Small volume, high-touch.
Subscription	Deferred	Deferred	Not modeled before meaningful scale.

CAC and Key Business Metrics

Metric	Pilot 1	Pilot 2	Production
CAC	\$10	\$16	\$22
Gross margin	35%	45%	55%
Monthly ARPU	\$1.5	\$2.5	\$4
Monthly churn	15%	12%	8%
LTV/CAC target	1.2x	2.0x	2.8x

Cash Flow and Runway

Phase Summary

Checkpoint	Ending Cash (USD)	Ending Cash (INR)	Approx. Runway Remaining	Trigger Threshold
Pre-Pilot (M5)	\$4,350	INR 361,050	0.7 months	Critical: must enter Pilot 1 immediately with investor conversation underway.
Pilot 1 (M7)	\$2,430	INR 201,690	0.3 months	Critical: investor close must happen by M7-M8 or company stalls.
Pilot 2 (M16)	\$39,510	INR 3,279,330	3.5 months	Trigger contingency if cash drops below 3 months of burn.
Production Readiness (M26)	\$21,880	INR 1,816,040	2.6 months	Trigger contingency if cash drops below 3 months of burn.

Monthly Cash Flow

Month	Phase	Founder Capital In	Investor In	Revenue In	Outflow	Net	Ending Cash
M1	Pre-Pilot	\$3,800	\$0	\$0	\$3,010	\$790	\$790
M2	Pre-Pilot	\$3,800	\$0	\$0	\$3,010	\$790	\$1,580
M3	Pre-Pilot	\$3,800	\$0	\$0	\$3,010	\$790	\$2,370
M4	Pre-Pilot	\$3,800	\$0	\$0	\$3,010	\$790	\$3,160
M5	Pre-Pilot	\$3,800	\$0	\$0	\$3,610	\$190	\$3,350
M6	Pilot 1	\$3,800	\$0	\$300	\$8,189	-\$4,089	-\$739
M7	Pilot 1	\$3,800	\$0	\$600	\$8,189	-\$3,789	-\$4,528
M8	Pilot 2	\$0	\$84,000	\$1,000	\$10,858	\$74,142	\$69,614
M9	Pilot 2	\$0	\$0	\$1,400	\$10,858	-\$9,458	\$60,156
M10	Pilot 2	\$0	\$0	\$1,800	\$10,858	-\$9,058	\$51,098
M11	Pilot 2	\$0	\$0	\$2,200	\$10,858	-\$8,658	\$42,440
M12	Pilot 2	\$0	\$0	\$2,600	\$10,858	-\$8,258	\$34,182

M13	Pilot 2	\$0	\$0	\$3,000	\$10,858	-\$7,858	\$26,324
M14	Pilot 2	\$0	\$0	\$3,400	\$10,858	-\$7,458	\$18,866
M15	Pilot 2	\$0	\$0	\$3,800	\$10,858	-\$7,058	\$11,808
M16	Pilot 2	\$0	\$0	\$4,200	\$10,858	-\$6,658	\$5,150
M17	Prod. Readiness	\$0	\$0	\$4,600	\$12,440	-\$7,840	-\$2,690
M18	Prod. Readiness	\$0	\$0	\$5,000	\$12,440	-\$7,440	-\$10,130
M19	Prod. Readiness	\$0	\$0	\$5,500	\$12,440	-\$6,940	-\$17,070
M20	Prod. Readiness	\$0	\$0	\$6,200	\$12,440	-\$6,240	-\$23,310
M21	Prod. Readiness	\$0	\$0	\$7,000	\$12,440	-\$5,440	-\$28,750
M22	Prod. Readiness	\$0	\$0	\$8,000	\$12,440	-\$4,440	-\$33,190
M23	Prod. Readiness	\$0	\$0	\$9,000	\$12,440	-\$3,440	-\$36,630
M24	Prod. Readiness	\$0	\$0	\$10,000	\$12,440	-\$2,440	-\$39,070
M25	Prod. Readiness	\$0	\$0	\$11,500	\$12,440	-\$940	-\$40,010
M26	Prod. Readiness	\$0	\$0	\$13,000	\$12,440	\$560	-\$39,450

Note on negative cash positions: The cash flow above shows that at the low end of the investor capital range (\$84,000), runway is exhausted by M17. The \$25,000 contingency buffer (included in total economic cost but not shown as a line item) and the upper investor range (\$103,000) partially address this. At the upper investor range, ending cash at M16 would be ~\$24,150 and at M26 would be ~-\$20,450. **This scenario requires either (a) revenue to exceed modeled projections, (b) investor capital at the high end of the range, or (c) a bridge round during Production Readiness.** This is the core reason S3-O1-I1 is rated "Avoid."

Totals (midpoint estimates):

- Founder capital in: \$26,600 / INR 2,207,800 (M1-M7 at ~\$3,800 x 7 months)
- Investor capital in: \$93,500 / INR 7,760,500 (midpoint of \$84,000-103,000)
- Cumulative revenue (M26): ~\$103,100
- Cumulative outflow (M26): ~\$118,500

ROI and Break-Even

Metric	Value
Total capital invested (founder + investor)	\$105,000-135,000 / INR 8,715,000-11,205,000
Total economic cost	\$105,000-135,000 / INR 8,715,000-11,205,000
Sweat-equity gap (economic cost minus cash invested)	Minimal — bootstrapped level means most economic cost is actual cash
Cumulative revenue at M26	~\$103,100 / INR 8,557,300
Monthly break-even (revenue >= outflow)	~M32-M40 (6-14 months post-Production)
Cumulative break-even (total revenue >= total capital)	~M34-M42 (8-16 months post-Production)
Time to ROI (revenue exceeds all economic cost)	~M40-M48 (14-22 months post-Production)

Fundraising, Valuation, and Dilution

Item	Value
Founder/self-funded capital before investor	\$21,000-32,000 / INR 1,743,000-2,656,000
Investor round size	\$84,000-103,000 / INR 6,972,000-8,549,000
Recommended structure	SAFE or milestone-based bridge note
Modeled pre-money valuation	\$420,000-515,000
Modeled post-money valuation	\$504,000-618,000
Modeled investor dilution from the 20% pool	16.7%

Ownership Pool	Before Round	After Modeled Round	Notes
Owner / founder pool	60%	56.0%	Retained majority control.
Founding + strategic pool	20%	18.7%	Shared pool for founding members and strategic partners.
Investor pool	20% reserved	16.7% issued / 3.3% remaining	Drawn from the dedicated investor pool with minor proportional dilution.

Key Risks

Risk	Likelihood	Impact	Mitigation / Trigger
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Investor does not close by M8	High	Critical	Runway is essentially zero by M7. Pre-negotiate term sheet during Pilot 1 (M5-M7). Must have backup plan (personal bridge loan, friends/family round).
Cash runway exhaustion in Production Readiness	Very High	Critical	Bootstrapped investor capital is insufficient to cover full Production Readiness at modeled costs. Requires either bridge financing, accelerated revenue, or scope reduction.
Team fatigue over 26-32 month timeline	High	High	Maintain clear phase gates. Celebrate milestones. Allow rest periods between phases.
Bootstrapped investor adds insufficient capital	High	High	\$84,000-103,000 is the bare minimum and may not reach Production. Must hit revenue targets by M20 or seek bridge.
Low founder skin-in-game perception	Medium	High	\$21,000-32,000 is low by investor standards. Offset with sweat equity narrative and demonstrated Pilot 1 commitment.
Manual ops overload	High	Medium	Add admin capacity before feature expansion. Track callback backlog as release gate.
Regulatory delay (Ethiopia / Kenya)	Medium	High	Continue partner-license model. Count friendly counsel at economic value.
Payment-rail mismatch	Medium	Medium	Use digital invoicing first; localize M-Pesa / Telebirr before scaling.
Security gaps from extended timeline	Medium	Medium	Maintain release freeze discipline. Do not skip hardening gates even under time pressure.

Recommendation

S3-O1-I1 is the **highest-risk combination** within Scenario 3. While the earlier investor entry (right after Pilot 1) reduces founder capital exposure to the lowest level in the entire 21-combination matrix (\$21,000-32,000), the double-bootstrapped budgets create a fundamentally fragile path. The investor capital of \$84,000-103,000 is insufficient to cover Pilot 2 and Production Readiness at even minimum staffing levels, as shown by the negative cash positions beginning at M17.

This path should be avoided. It suffers from three compounding problems:

1. **Insufficient investor capital** — bootstrapped investor funding cannot bridge the gap from Pilot 2 through Production at viable staffing levels
2. **Zero runway margin** — the M6-M7 transition has less than 1 month of cash, meaning any investor delay is immediately fatal
3. **Revenue dependency** — the model requires revenue to ramp faster than historical comparables for similar-stage health-tech platforms to avoid cash-out during Production Readiness

If this is the only option available:

1. The investor must commit at the high end of the range (\$103,000) with funds available by M8
2. Revenue targets in Pilot 2 must be treated as hard gates, not aspirational
3. A contingency bridge round (\$15,000-25,000) should be pre-arranged for M16-M18
4. Production Readiness scope must be aggressively compressed to 8 months maximum

Consider S3-O1-I2 or S3-O1-I3 for meaningful risk reduction — the same low founder capital exposure with substantially more investor capital.

Assumption Notes

- Android is modeled from the external quote: 480 total hours at \$15/hr average across the mobile team.
- Remaining web work is normalized to 150%-160% of Android effort, with an expected case of 745 hours.
- Salary benchmarks for employed coverage: INR 60,000 for doctors (\$720/mo), INR 18,000 for admin ops (\$217/mo), INR 25,000 for technical support (\$301/mo).
- Our L1 (bootstrapped) spend is modeled at ~~40% of ideal team utilization, yielding ~\$3,800/mo founder contribution pre-investor~~ (\$2,900 team + ~\$900 ops/infra/overhead). Lower than S2-O1-I1's \$8,900/mo because the self-funded window is shorter (7 months vs. 9 months) and must be sized to realistic bootstrapped constraints.
- Investor L1 (bootstrapped) contribution sized at \$84,000-103,000, representing bare-minimum capital to attempt Production at constrained pace.
- Pre-Pilot duration compressed to 5 months (vs. 6 in S2) reflecting the urgency to reach investor conversation.
- All INR conversions at 1 USD = 83 INR per params.md.
- FX buffers of 10-15% applied to ETB and KES revenue projections.

End of S3-O1-I1 scenario. All assumptions trace to params.md v2.